

FLINT BISHOP, MI, USA

WMO: 726370

Lat: 42.967N

Lon: 83.749W

Elev: 770

StdP: 14.29

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14826

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.1	(c) 3.9	(d) -8.3	(e) 3.6	(f) 0.7	(g) -3.4	(h) 4.7	(i) 5.3	(j) 26.3	(k) 26.6	(l) 24.1	(m) 23.1	(n) 6.4	(o) 270	(p) 0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	89.9	73.1	87.1	71.9	84.4	70.2	75.9	85.5	74.2	83.3	72.6	81.1	10.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	125.6	81.9	71.2	118.4	79.4	69.6	112.0	77.5	39.9	85.7	38.1	83.6	36.6	81.1	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	20.2	18.2	-9.4	94.2	6.5	2.8	-14.1	96.2	-17.9	97.9	-21.6	99.4	-26.3	101.5
			-9.9	79.2	6.4	2.0	-14.5	80.6	-18.2	81.8	-21.8	82.9	-26.5	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	23.4	25.5	34.8	46.9	58.3	68.0	71.7	70.0	62.9	51.2	39.6	29.4
	DBStd	19.28	11.57	11.09	11.40	9.22	8.63	7.00	5.91	5.67	7.64	8.75	9.34	9.93
	HDD50	3261	825	688	493	165	20	0	0	0	3	93	335	639
	HDD65	6636	1290	1107	939	547	247	48	11	18	128	436	763	1102
	CDD50	2749	1	1	21	71	276	541	673	620	391	130	22	2
	CDD65	647	0	0	1	3	39	138	218	174	66	8	0	0
	CDH74	6076	0	0	11	50	411	1328	2129	1469	610	67	1	0
	CDH80	1865	0	0	2	6	102	430	731	420	163	11	0	0
	WSAvg	8.7	10.0	10.0	9.8	10.2	8.7	7.5	7.0	6.6	7.1	8.6	9.4	9.5
	PrecAvg	30.7	1.7	1.5	2.0	3.0	3.1	3.1	3.0	3.3	3.2	2.4	2.4	2.0
Precipitation	PrecMax	45.4	4.0	4.0	4.3	6.5	8.2	6.6	9.4	11.0	10.9	6.6	5.7	4.6
	PrecMin	18.1	0.3	0.2	0.6	0.7	0.4	0.6	0.7	0.3	0.3	0.3	0.6	0.4
	PrecStd	5.5	0.9	0.9	0.9	1.4	1.8	1.5	1.7	1.8	2.1	1.1	1.2	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	58.1	74.0	79.8	86.9	91.9	94.2	92.2	89.6	80.9	68.6	60.2
		MCWB	53.0	51.7	61.5	61.8	69.3	73.1	75.5	74.0	72.4	67.8	57.8	55.7
	2%	DB	49.7	51.0	64.8	73.6	82.8	88.4	90.2	87.7	84.5	74.7	62.3	52.5
		MCWB	46.3	46.7	55.3	58.7	67.2	72.0	73.7	73.0	69.4	63.0	54.9	48.7
	5%	DB	43.7	45.3	58.2	68.5	79.0	85.1	87.1	84.4	80.8	70.7	58.8	47.5
		MCWB	40.3	40.6	51.0	55.6	65.5	70.3	72.3	70.9	67.9	61.6	52.5	43.8
	10%	DB	38.4	39.9	51.9	63.4	74.3	81.6	84.1	81.8	77.1	66.3	54.3	42.5
		MCWB	36.0	36.5	45.2	52.7	62.6	68.2	70.4	69.5	65.8	58.3	48.7	39.0
	0.4%	WB	53.3	53.9	62.4	66.2	72.4	76.6	79.1	77.5	75.3	69.9	60.3	56.0
		MCDB	55.3	56.5	72.1	76.1	82.9	87.7	90.1	87.3	85.4	78.4	65.0	59.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.0	46.8	57.0	61.2	69.5	74.2	76.5	75.2	72.2	66.2	56.9	49.8
		MCDB	49.0	50.4	63.2	69.4	79.0	84.9	86.2	83.2	80.4	71.9	61.1	52.4
	5%	WB	40.5	40.8	51.6	57.8	67.3	72.3	74.5	73.3	70.1	62.5	53.0	44.1
		MCDB	43.3	44.4	57.1	66.5	75.9	82.1	83.7	81.1	77.1	68.8	57.8	47.1
	10%	WB	36.2	36.7	45.6	53.7	64.2	70.4	72.5	71.4	67.9	59.4	49.0	39.1
		MCDB	38.4	39.8	51.2	62.6	73.0	78.7	81.0	79.0	75.1	65.1	53.9	41.9
Mean Daily Temperature Range	5% DB	MDBR	13.5	15.1	17.4	20.5	21.1	20.8	21.1	20.1	21.3	18.4	14.7	11.9
		MCDBR	17.2	19.4	24.7	27.4	24.9	23.5	23.1	22.4	24.8	24.2	20.1	16.7
		MCWBR	15.4	16.1	16.7	16.3	13.4	11.2	10.7	10.5	12.1	14.5	14.8	14.1
	5% WB	MCDBR	16.6	18.1	22.4	24.2	22.0	20.6	20.1	18.9	20.1	20.9	18.5	16.1
		MCWBR	15.5	15.8	16.6	16.7	13.4	11.2	10.8	10.2	11.4	14.2	15.3	14.4
Clear-Sky Solar Irradiance	taub		0.296	0.319	0.341	0.380	0.405	0.430	0.418	0.413	0.380	0.348	0.316	0.295
	taud		2.360	2.286	2.360	2.276	2.260	2.219	2.267	2.275	2.356	2.436	2.478	2.414
	Ebn at Noon		266	277	285	281	276	268	270	267	269	264	258	256
	Edh at Noon		25	32	34	40	42	44	42	40	34	28	23	22
All-Sky Solar Radiation	RadAvg		466	745	1128	1454	1701	1916	1931	1655	1328	847	521	367
	RadStd		44	95	117	128	149	114	92	68	124	79	67	31

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page